# From Retinal Flow to Predictive Visual Fields: Image-Native Language in Continuous State and Pixel Space

Imagized Language Model Project

August 2026

**Abstract**

Most language models represent language through symbolic tokens and treat visible writing as an auxiliary modality. We study a stricter object: an independent model whose inputs, state supervision, probability model, and primary outputs are continuous images of writing. Earlier Retinal Flow Language Models (RFLM) read ordered $32 \times 32$ image fixations, integrate them recurrently, write the next fixation by pixel flow, and reread generated ink; their monolithic writer remains unreadable. We therefore implement a Predictive Visual Field (PVF) that factorizes visual language from visual actuation. A frozen convolutional retina maps writing images to the unit sphere, a causal field integrates image history, a deterministic continuous proposal predicts a low-variance next visual state, and an intrinsic hyperspherical flow models alternatives. V16 adds residual multiscale causal memory with dilated local fields and global causal attention. The 16.47M-parameter student contains no tokenizer, strings after rasterization, Unicode IDs, OCR, character labels, output vocabulary, discrete visual codebook, candidate classifier, or external language model. On a frozen 512-form, four-view Chinese benchmark, selected V16 proposal top-1 is 6.26%, exceeding last-image-only (3.97%), unigram (1.73%), and random dynamics (0.22%); full history adds +0.0773 normalized target log-probability. The stochastic field reaches 3.69%, versus 3.24% last-only, and sampled context cosine gain is +0.0795. The model trains on one RTX 4090 with 1.479 GiB peak allocated CUDA memory. These results establish causal visual-state learning under a strict image-only boundary, but remain below a symbolic bigram (13.14%). We then train a separate 5.73M-parameter V17 visual actuator from continuous retinal state and style image to ink pixels. On one untouched 512-example evaluation, correct-state visual identity is 58.59% versus 0.98% after shuffling only intended states; target cosine is 0.713 versus 0.086. However, pixel F1 is 0.438, below the fixed 0.500 gate, and human review rejects readability. Thus a small continuous visual state causally controls generated writing on one consumer GPU, while exact stroke topology, general language, historical question answering, and efficiency over text models remain unproved.

## 1 Motivation

Human readers do not see byte-pair tokens. They see pages, lines, strokes, spacing, damage, font variation, and script-specific visual forms. Current language models are extremely capable, but their central representation is normally a sequence of text tokens. Multimodal models accept images, and modern APIs also expose image-generation endpoints. For example, OpenAI documents APIs for processing images as input and generating images as output, and Google's Gemini API documentation describes image generation from text and image prompts [4, 5, 6]. These systems are important baselines, but they do not by themselves define a language model whose primary language substrate is the visible page.

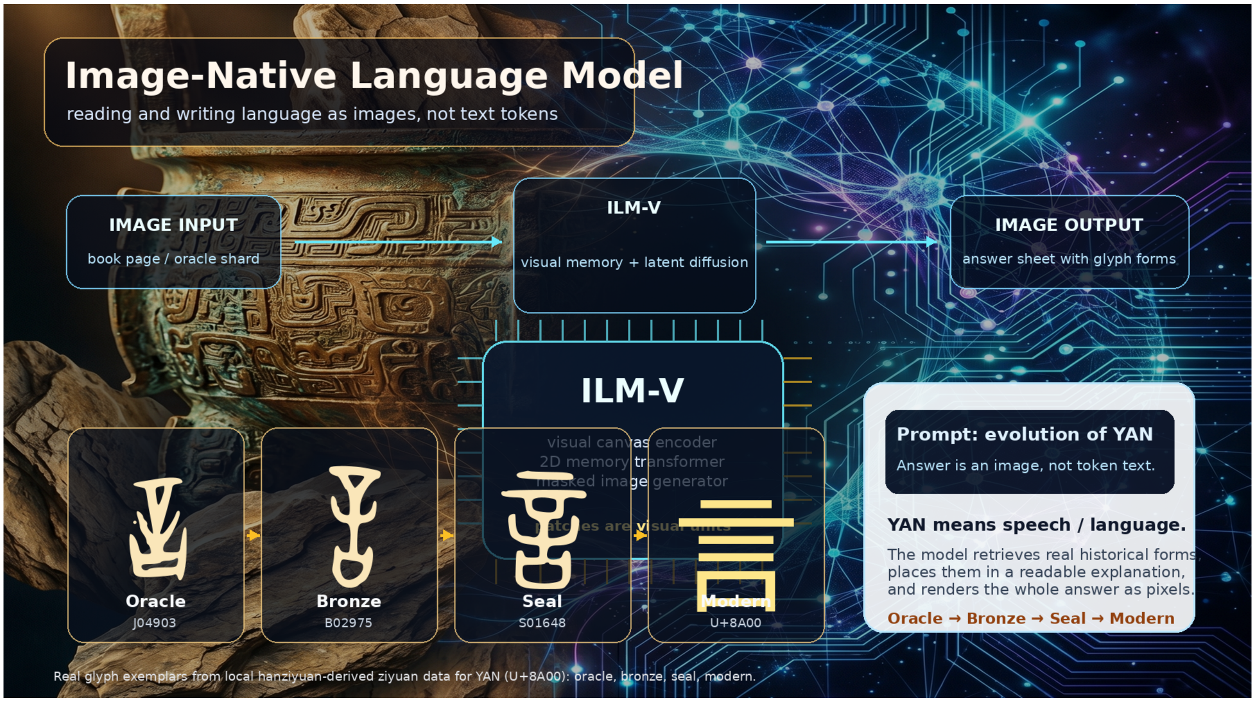


Figure 1: Image-native language modeling concept figure generated with an AgInTi background and overlaid with real local ziyuan glyph exemplars for `YAN` (U+8A00). The answer is a rendered image containing both modern explanation and historical forms.

The Imagized Language Model (ILM) goal is different:

$$\text{image input} \rightarrow \text{visual language state} \rightarrow \text{image output}.$$

The output should itself be a readable answer image. For a query such as "explain the evolution of character yan", the system should produce a page containing modern explanatory writing, classical-style Chinese, and oracle, bronze, seal, and modern forms. Typed text can still be accepted by a user interface, but it is rasterized into an image prompt before entering the core model. Conversely, the model's answer is a page image first; OCR or transcription is an optional downstream extraction step.

## 2 Prior Art and Gap

OCR-free document models such as Donut and screenshot parsing models such as Pix2Struct show that document images can be understood without an explicit OCR pipeline. PIXEL reconstructs masked rendered-text patches and demonstrates cross-script visual transfer [10]; PIXAR is a genuinely generative pixel-space autoregressive language model and identifies noisy maximum-likelihood text images as a central difficulty [11]; PixelGPT studies autoregressive visual and textual pretraining [12]. These are the closest model-level precedents. Token-free language models such as CANINE, ByT5, and MEGABYTE remove subwords but retain symbolic bytes or characters. Discrete diffusion language models show that generation need not be left-to-right, but usually retain a linguistic vocabulary.

I-JEPA shows that predicting target image representations from visual context can learn semantic visual fields without reconstructing every pixel [26]. Flow matching trains continuous generative

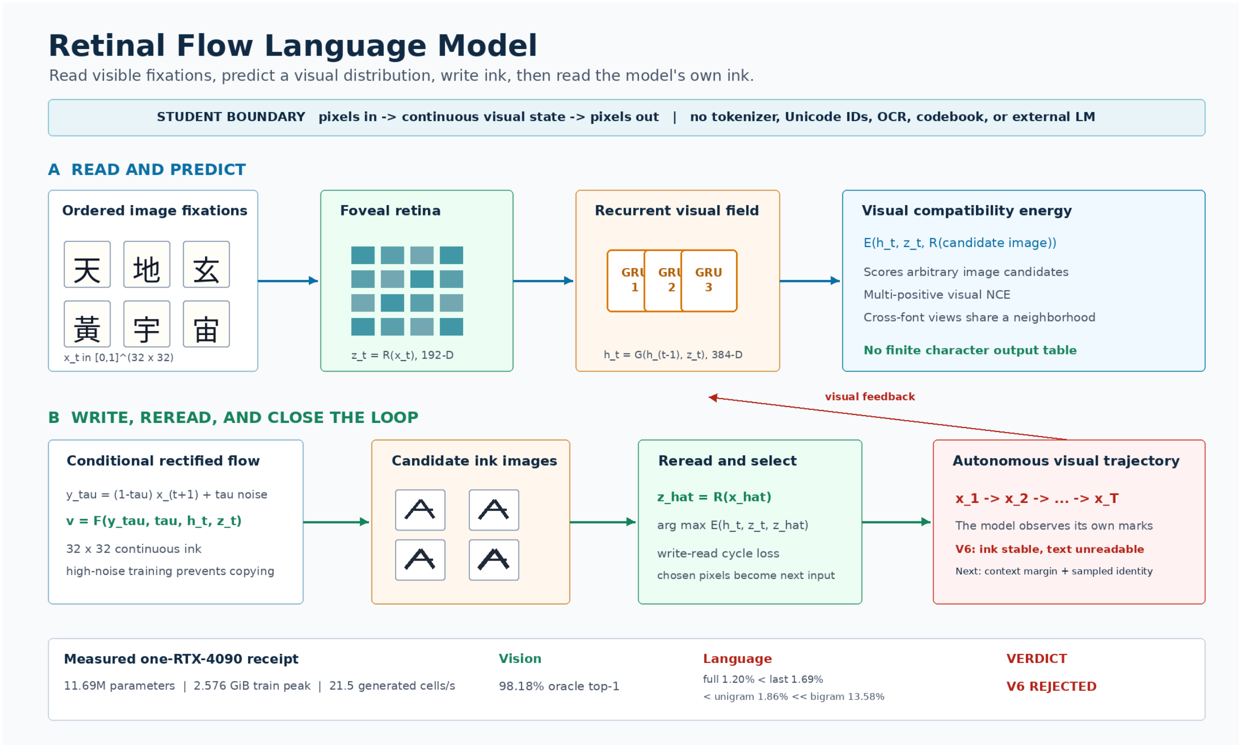


Figure 2: The implemented Retinal Flow Language Model. Training joins visual next-state energy, conditional rectified flow, a write–read cycle, V6 model-induced visual rollouts, and V7 calibrated context and sampled-image identity. Autonomous inference samples candidate ink images, rereads them, selects by visual energy, and feeds the selected pixels back.

paths by vector-field regression and supports efficient numerical sampling [23]. Human reading is also sequential and foveated: Chinese readers make saccades and can use both character- and word-related targeting cues [33]. These observations motivate RFLM, but they do not imply biological equivalence.

The remaining gap is not simply "pixels instead of tokens." It is a system whose predictive random variables are continuous writing images, whose memory is learned from ordered visual observations, whose writer produces new pixels, and whose own output distribution is present during evaluation. Historical evidence adds a separate constraint: synthesized marks must never be presented as attested source glyphs.

# 3 Architecture

Figure 2 gives the strict visual contract. For fixation images $x_t \in [0,1]^{H\times W}$, RFLM factors a visual sequence as

$$p_\Theta(x_{1:T}) = p(x_1)\prod_{t=1}^{T-1} p_\Theta(x_{t+1} \mid x_{\le t}).$$

The random variables are continuous image fields rather than entries in a linguistic or visual vocabulary.

**Foveal retina.** A convolutional reader maps each $32 \times 32$ image to a normalized 192-dimensional visual state:

$$z_t = R_\theta(x_t)/\|R_\theta(x_t)\|_2.$$

Independent font renderings are aligned with visual contrastive, invariance, and variance objectives. The target retina $R_{\bar{\theta}}$ is an exponential moving average of the online retina. Font coverage is checked from font cmap tables so missing-glyph boxes cannot become a spurious class.

**Recurrent visual field and energy.** A three-layer GRU integrates the fixation history,

$$h_t = G_\phi(h_{t-1}, z_t), \qquad h_t \in \mathbb{R}^{384}.$$

An energy function scores any candidate image $c_j$ after rereading it:

$$s_{t,j} = E_\psi(h_t, z_t, R_{\bar{\theta}}(c_j)).$$

Multi-positive visual NCE admits multiple near-identical image views as valid targets. Unlike a softmax language head, this energy is not tied to a finite character inventory.

**Conditional ink flow.** For clean target $x_{t+1}$, noise $\epsilon$, and $\tau \in [0, 1]$, define

$$y_\tau = (1-\tau)x_{t+1} + \tau\epsilon, \qquad u_\tau = \epsilon - x_{t+1}.$$

The pixel writer predicts the path velocity

$$\hat{u}_\omega = F_\omega(y_\tau, \tau, h_t, z_t), \qquad \mathcal{L}_{\text{flow}} = \|\hat{u}_\omega - u_\tau\|_2^2.$$

Training emphasizes high-noise times, where target pixels cannot simply be copied, and weights sparse strokes and the predicted clean endpoint.

**Write–read cycle.** The one-step endpoint $\hat{x}_0 = y_\tau - \tau\hat{u}_\omega$ is reread through the frozen target retina. A visual NCE cycle objective aligns it to independent target renderings:

$$\mathcal{L}_{\text{cycle}} = \text{NCE}\Big(R_{\bar{\theta}}(\hat{x}_0), R_{\bar{\theta}}(x_{t+1}^{(1)}), R_{\bar{\theta}}(x_{t+1}^{(2)})\Big).$$

The complete loss combines flow, energy, cross-render invariance, retina contrast, endpoint, stroke, and cycle terms. At inference, the writer samples a candidate set $\mathcal{C}_t$ and selects

$$\hat{x}_{t+1} = \arg\max_{c \in \mathcal{C}_t} E_\psi(h_t, z_t, R_{\bar{\theta}}(c)).$$

The selected pixels become the next observation. This closes the visual loop and turns autonomous stability into a mandatory model property rather than a post-processing concern.

**Model-induced visual rollout.** V6 invokes the same sampler and energy reranker during training. From a clean prefix it generates two short-flow candidates, selects and detaches one bitmap, rereads it with the online retina, and advances the recurrent state. For two rollout steps it minimizes

$$\mathcal{L}_{\text{roll}} = 0.15\mathcal{L}_{\text{state}} + 0.35\mathcal{L}_{\text{energy}}^{\text{roll}} + 0.30\mathcal{L}_{\text{recovery}}.$$

The terms align clean and induced recurrent states, predict the real next image from the induced state, and recover clean next pixels after generated feedback. All targets remain continuous image fields. Candidate selection is stop-gradient, which bounds memory while exposing the trainable state and writer to deployed pixels.

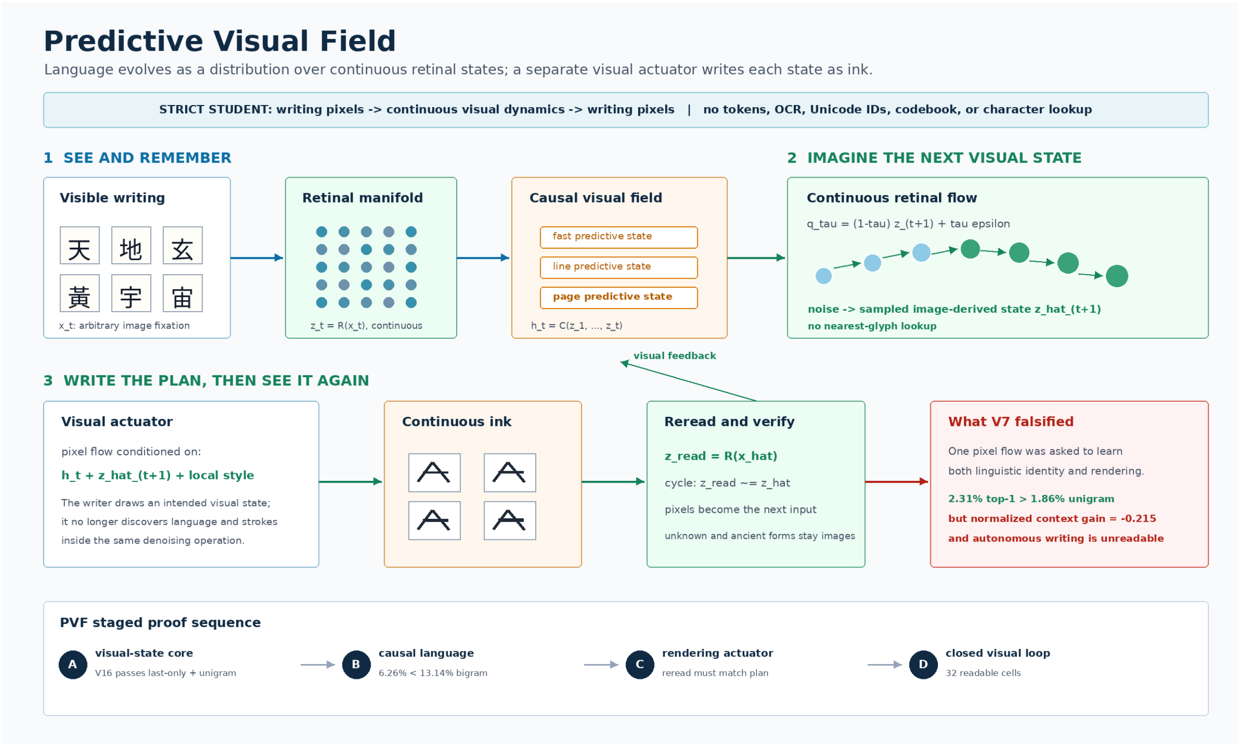


Figure 3: Predictive Visual Field factorization. A causal image field predicts a low-variance continuous visual proposal and a stochastic distribution over image-derived retinal states. A later separate actuator renders a selected state and the retina rereads its pixels.

**Calibrated context and sampled identity.** V7 corrects two training/evaluation mismatches. Raw target-energy differences can be increased by changing the scale or all competing scores, so V7 uses the normalized visual target log probability

$$\ell(y \mid h, \mathcal{C}) = s(y \mid h) - \log \sum_{c \in \mathcal{C}} \exp s(c \mid h)$$

and minimizes

$$\mathcal{L}_{\text{ctx}} = \max\{0, m - \ell(y \mid h_{\text{full}}, \mathcal{C}) + \ell(y \mid h_{\text{last}}, \mathcal{C})\}.$$

The last-only branch is detached. The training-only set $\mathcal{C}$ contains independent image renderings; positives are inferred by retinal similarity. Offline strings select the curriculum but anchor IDs and target indices do not enter the student, the pixels are disjoint from evaluation views, and the anchors are not deployed. V7 also differentiates through a two-step numerical pixel-flow endpoint,

$$\mathcal{L}_{\text{sample}} = \text{NCE}\left(R_{\bar{\theta}}(\hat{x}_0^{K=2}), \{R_{\bar{\theta}}(y^{(v)})\}_{v=1}^{V}\right).$$

This trains the sampled image path rather than only a one-step endpoint proxy.

# 4 Predictive Visual Field

Figure 3 separates linguistic uncertainty from stroke rendering. For target state $z_{t+1} = R(x_{t+1}) \in \mathbb{S}^{d-1}$, a deterministic proposal predicts

$$\mu_t = \frac{P_\psi([h_t, z_t])}{\|P_\psi([h_t, z_t])\|_2}.$$

The proposal is a continuous image-derived state, not a class index. It is trained with geodesic alignment, image contrast, and full-history advantage against a detached last-image branch.

In parallel, choose random unit source $\epsilon$, let $q_\tau$ follow the shortest spherical geodesic from $z_{t+1}$ to $\epsilon$, and let $u_\tau$ be its tangent velocity. The conditional field learns

$$\mathcal{L}_{\text{state-flow}} = \mathbb{E}\left\|v_\eta(q_\tau, \tau, [h_t, z_t]) - u_\tau\right\|_2^2.$$

The loss is full tangent-vector energy, not coordinate-mean MSE. Sampling integrates backward with the spherical exponential map,

$$q_{k-1} = \mathrm{Exp}_{q_k}\left(-\Delta\tau\, v_\eta(q_k, \tau_k, [h_t, z_t])\right),$$

yielding intended next visual states $\hat{z}_{t+1}$, not glyph IDs. Candidate images are used only by the evaluator: it rereads them with the frozen retina and computes kernel density against continuous samples. There is no learned 512-way output layer.

The training-only multiview anchor curriculum contains image tensors. Positives are inferred from retinal cosine similarity; labels and target indices do not enter the student, and the bank is not deployed. This visual contrast is necessary because geodesic endpoint alignment alone can reward a visual mean without separating language alternatives.

V16 augments the pretrained recurrent trajectory $h^R_{1:t}$ with three residual causal memory blocks. Each block combines a left-padded depthwise temporal field at dilation 1, 2, or 4, global causal attention, and a gated feed-forward residual. For memory $M_\phi$, the corrected state is

$$h_t^{V16} = h_t^R + \sigma(g) W_o \,\mathrm{LN}\{M_\phi(h^R_{1:t}, z_{1:t})\}.$$

The per-channel gate is initialized at $g = -4$, preserving the trained V15 function at the start of the upgrade. The selected checkpoint has mean $\sigma(g) = 0.018023$; therefore the frozen result is a bounded memory pilot, not evidence that a large residual correction is necessary.

V17 implements a separate visual actuator that writes

$$\hat{x}_{t+1} = W_\omega(\epsilon_x, h_t, \hat{z}_{t+1}, R_\theta(x_t))$$

and is trained with pixel flow plus a reread constraint

$$\mathcal{L}_{\text{act}} = \mathcal{L}_{\text{pixel-flow}} + \lambda\left[1 - \cos\{R_{\bar{\theta}}(\hat{x}_{t+1}), z_{t+1}\}\right].$$

No nearest-glyph projection, character classifier, or token unembedding is permitted. In V17, $\hat{z}_{t+1}$ is supplied by a different-font image of the target form so actuation is isolated from language prediction. This differs from Embedded Language Flows, which demonstrates effective continuous flow dynamics but encodes and finally unembeds discrete tokens [29]. It combines the next-fixation representation objective of recurrent JEPA [27] with the prediction/generation separation explored by D-JEPA [28], and uses intrinsic flow matching on a Riemannian manifold [24, 25], while retaining an image-derived language state.

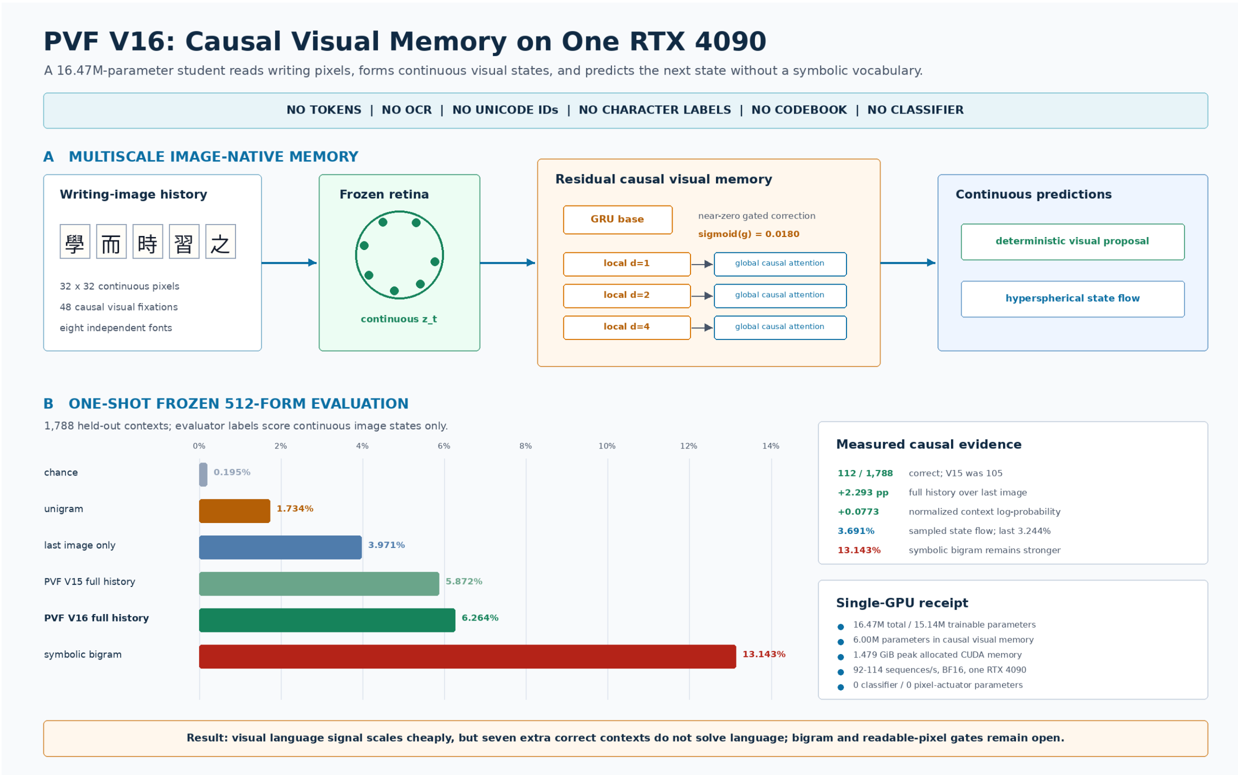


Figure 4: Measured PVF V16 memory result. Both deterministic proposal and stochastic hyperspherical field use full image history and beat random, last-only, and unigram controls. V16 improves proposal top-1 by seven frozen contexts over V15, but remains below the symbolic bigram and has no pixel actuator.

## 5 Isolated Visual Actuation

The V17 actuator tests whether an image-derived continuous state can control new pixels without a glyph table. A frozen retina produces $z = R(x^{\text{semantic}}) \in \mathbb{S}^{191}$ from one font view, and a convolutional encoder produces style vector $s = E_\omega(x^{\text{style}})$ from a different character in the desired target render. A 5.73M-parameter conditioned U-Net integrates rectified flow from Gaussian noise to a $32 \times 32$ ink image. The target image enters losses only; the spatial condition to the writer is a constant blank plane.

Training combines stroke-weighted flow matching, an endpoint loss, retinal cosine, multi-positive retinal contrast, and gradients through a two-step deployed sampler. At evaluation, correct and shuffled branches receive identical style images and initial noise. Only the intended state is rolled within the batch. The global evaluator retrieves among all 512 generated examples, with duplicate positives inferred from image-state similarity rather than labels.

V17 was selected at step 1,600 by a preregistered development rule and queried the frozen record split once. Correct-state top-1 is 58.594%, versus 0.977% shuffled; target cosine is 0.71301 versus 0.08612; and style-copy cosine is only 0.05858. These measurements reject style copying as the explanation for the causal signal. Pixel F1 is 0.43849 versus 0.28001 shuffled, failing the required 0.50, and human review rejects most outputs as unreadable. Training takes 339.67 seconds and peaks at 1.588 GiB allocated CUDA memory on one RTX 4090.

The result identifies a topology gap rather than an identity-control gap. V18 therefore decodes

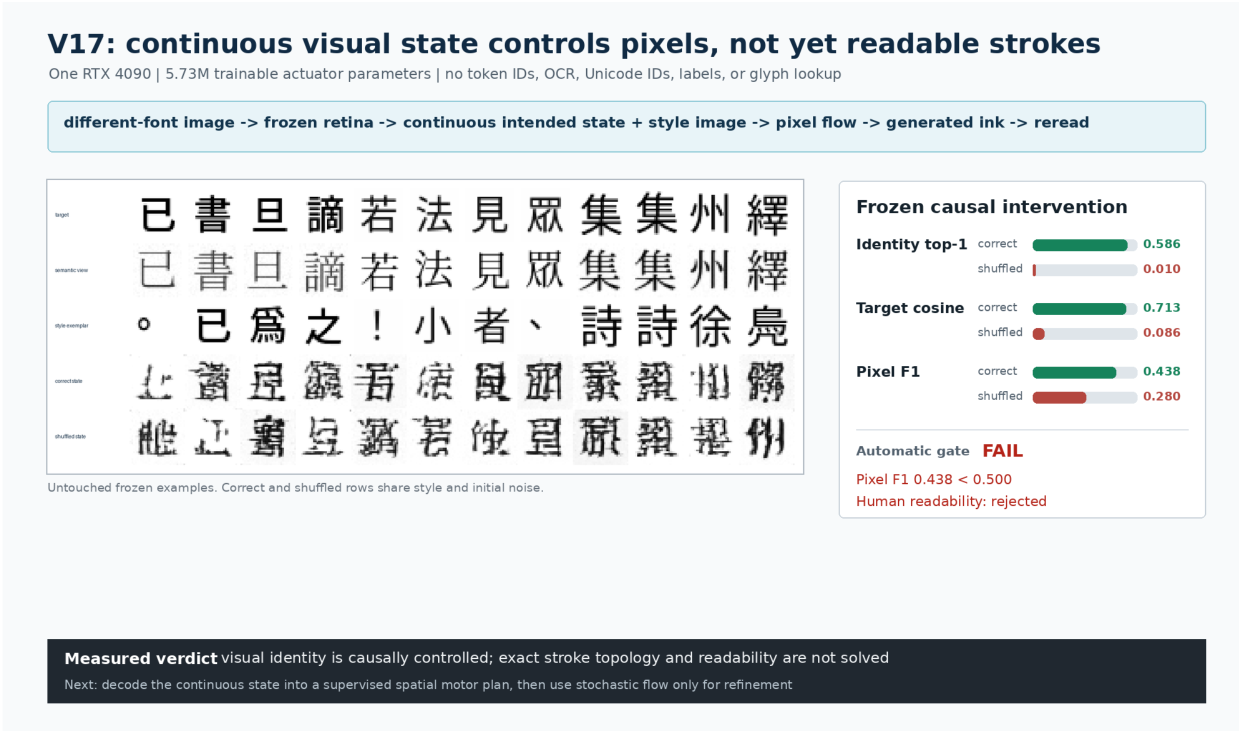


Figure 5: Measured V17 frozen result. Continuous intended state strongly controls recovered visual identity under a same-noise shuffled-state intervention, but pixel F1 and human readability fail. The correct-state row contains state-dependent pseudo-characters, not reliably readable target forms.

state and style into a directly supervised spatial visual motor plan,

$$p_\omega = D_\omega(z, E_s(x^{\text{style}})) \in [0,1]^{H\times W},$$

and reserves stochastic flow for residual style and surface variation. The plan remains continuous and image-derived; it is not copied target ink, an ID lookup, or a discrete visual codebook.

# 6 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

| Item | Count |
|---|---|
| Characters | 9,055 |
| Glyph records | 84,642 |
| Oracle | 32,216 |
| Bronze | 23,500 |
| Liushutong | 21,923 |
| Seal | 6,996 |

This supports future evidence-bearing historical answer pages but was not used as a character-label channel in the RFLM run. The language trajectory corpus is a 7,017-record manifest built from 16 public-domain Chinese Wikisource book packages. The contribution layer is recorded as CC BY-SA, and the manifest stores source artifacts and rights. Strings exist only in the offline manifest and renderer; model batches contain image tensors.

Eight Noto Sans/Serif CJK fonts provide independent visual views. The run stores the exact font paths, byte counts, and SHA-256 hashes in checkpoints and receipts. Cmap intersection filtering leaves 8,282 supported visible characters; 107 unsupported unique forms account for only 0.0074% of visible corpus instances and are skipped instead of rendered as missing-glyph boxes.

# 7 Training Objectives

Each 48-fixation page supplies eight sampled causal positions, making the language objective denser than one endpoint per page. The complete training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{flow}} + \lambda_e \mathcal{L}_{\text{energy}} + \lambda_r \mathcal{L}_{\text{retina}} + \lambda_v \mathcal{L}_{\text{view}} + \lambda_p \mathcal{L}_{\text{endpoint}} + \lambda_s \mathcal{L}_{\text{stroke}} + \lambda_c \mathcal{L}_{\text{cycle}} + \rho(t) \mathcal{L}_{\text{roll}}.$$

After step 5,200, V7 adds a ramped correction

$$\mathcal{L}_{\text{V7}} = \mathcal{L} + \kappa(t)\left(0.5\mathcal{L}_{\text{ctx}} + 0.2\mathcal{L}_{\text{sample}}\right).$$

The energy term uses 512 cross-render candidate views per batch in the final run. Visual duplicate positives are discovered by image similarity rather than labels. The view term aligns independent renderings in both retinal and candidate-projection spaces. Endpoint and stroke terms improve sparse-ink geometry. The cycle term is weighted toward high-noise examples, where a writer must use recurrent context rather than visible target pixels.

For V6, $\rho(t)$ ramps from zero to one over 400 updates. A rollout subset of eight sequences uses two generated steps, two candidates, and two flow steps per candidate. This follows the distribution-shift lesson of scheduled sampling and DAgger [31, 30]; it remains image-only because the visited states, selected observations, and recovery targets are pixels and continuous visual fields. The final rollout state cosine reaches 0.961, but selected-image target cosine is only 0.103. This distinction predicts the measured outcome: state recovery becomes robust while sampled visual identity remains weak.

V7 resumes V6 for 800 updates with a 512-form, four-view image-anchor curriculum and a two-step differentiable endpoint on four examples per batch. The anchor retina is refreshed every 200 updates. The selected step 5,800 checkpoint is chosen from preregistered validation checkpoints because it has the strongest combination of independent-anchor context gain (+0.1798) and sampled pixel F1 (0.3636). The continuation adds approximately 261 seconds of training and peaks near 3.0 GiB allocated VRAM on one RTX 4090.

## 7.1 PVF state objectives

V8–V16 reuse the proven retina but remove the pixel writer from the state ablation. The causal GRU, proposal, and hyperspherical field are trainable. For proposal $\mu$, target image state $z^+$, and image-derived candidate states $\{z_j\}$, visual identity uses

$$\mathcal{L}_{\text{proposal-NCE}} = -\log \frac{\sum_{j \in \mathcal{P}} \exp\{s\mu^\top z_j\}}{\sum_j \exp\{s\mu^\top z_j\}},$$

where positives $\mathcal{P}$ are inferred by frozen retinal similarity. A detached last-image condition supplies the calibrated context objective

$$\mathcal{L}_{\text{proposal-ctx}} = \max\{0, m - \ell(z^+ \mid h_{\text{full}}) + \ell(z^+ \mid h_{\text{last}})\}.$$

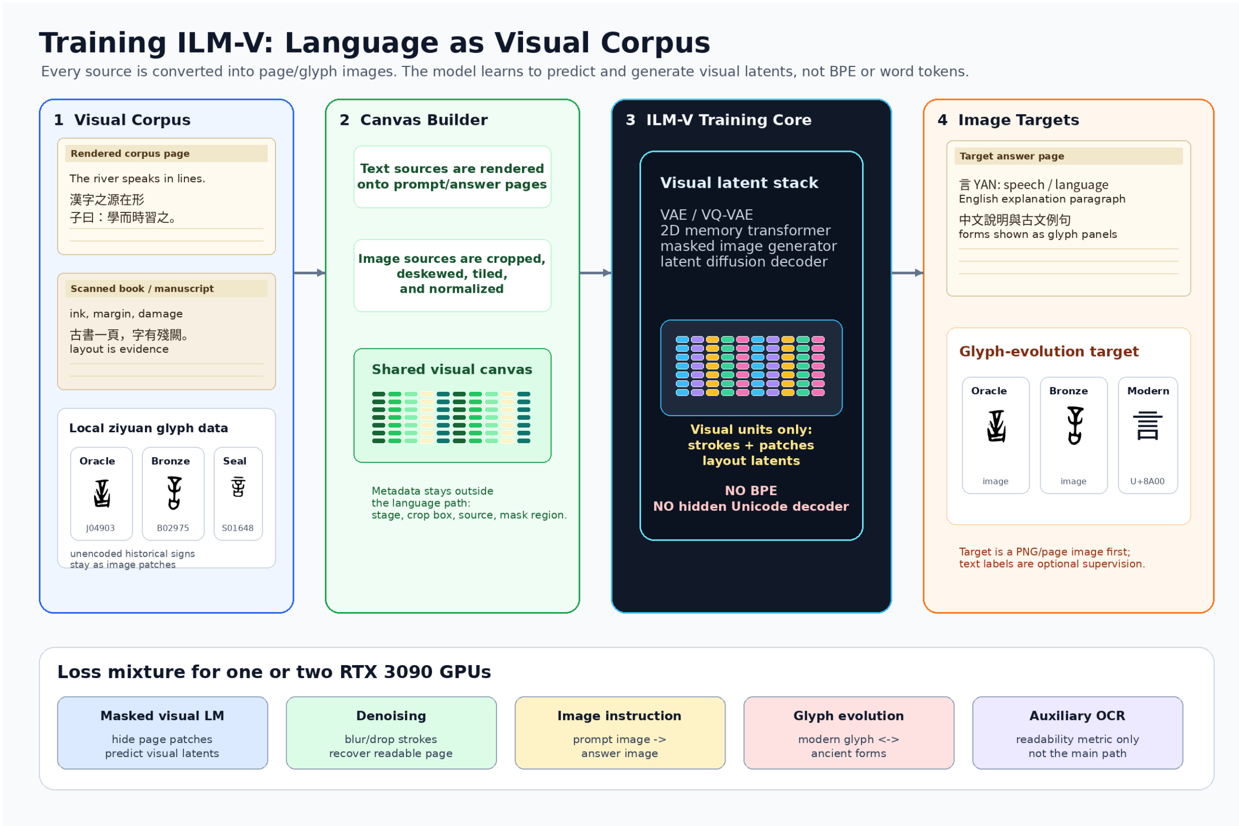


Figure 6: Training pipeline for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks are all converted into visual canvases. The training target is visual latent prediction and image generation, not BPE-token prediction.

Equivalent identity and context losses are applied to states numerically sampled from the flow. Direct sampled geodesic loss prevents inference-path drift, while contrastive loss prevents collapse to the retinal mean.

V15 has 10,470,273 total parameters, 9,137,345 trainable parameters, and zero classifier or pixel-actuator parameters. It trains in BF16 with 48-image sequences and peaks at 1.181 GiB allocated CUDA memory. The full V14–V15 trajectory records 1,895.24 seconds on one RTX 4090. Step 2,000 was selected on the development image bank before running the frozen evaluator.

V16 resumes selected V15 step 2,000, adds 6,001,536 causal-memory parameters, and resets the optimizer with a new 100-step warmup. It has 16,471,809 total parameters, 15,138,881 trainable parameters, and zero classifier or actuator parameters. The 1,200-update continuation records 2,047.63 seconds, about 92–114 sequences per second, and 1.479 GiB peak allocated CUDA memory. Step 2,200 was selected before the frozen evaluator by maximum development proposal top-1 subject to full-context over last-only, positive normalized context gain, and target cosine above 0.10.

## 8 Image-First Training and Inference

Figure 6 makes the training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and

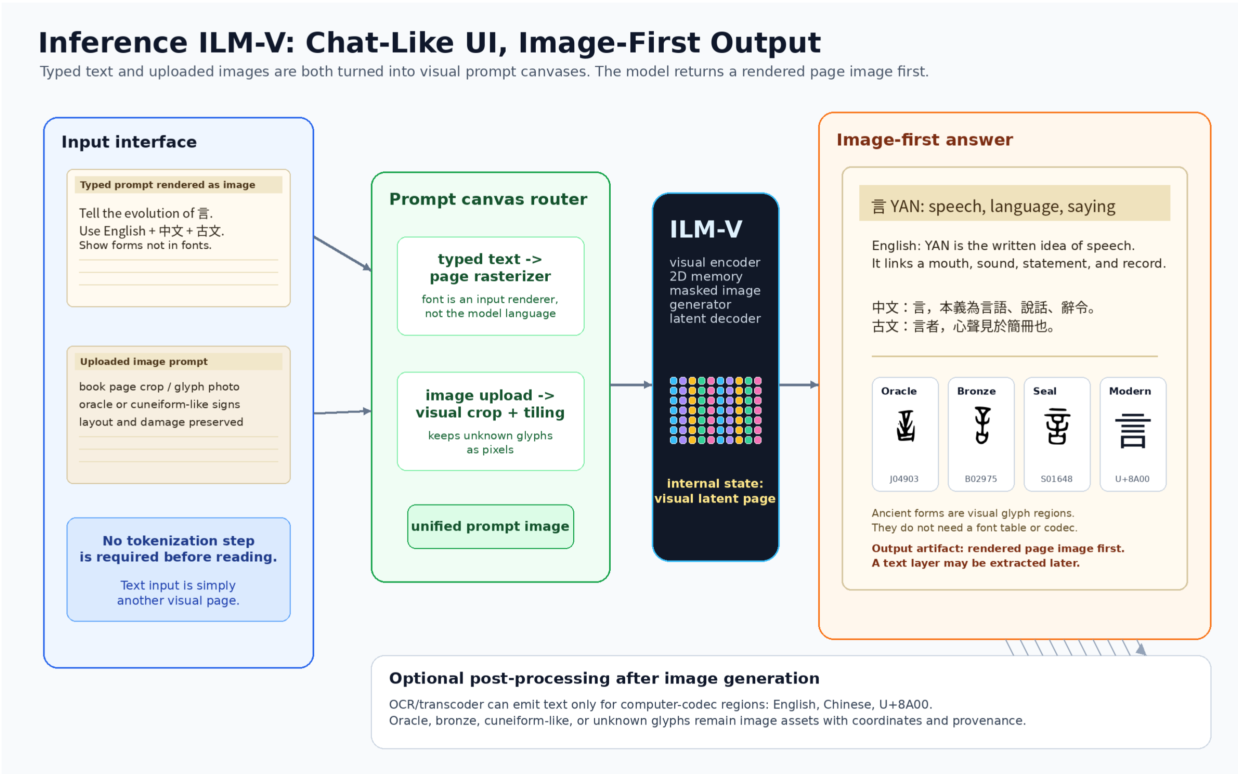


Figure 7: Inference pipeline for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The primary output is a rendered page image. OCR or transcoding is optional and only applies to output regions that exist in computer codecs.

stage labels may remain in manifests, but the model path should learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

Figure 7 shows the user-facing path. A typed prompt is converted into a visual prompt canvas, and an uploaded image is normalized into the same visual space. The model returns a page-like image containing modern language and historical glyph regions. A later OCR/transcoder can attach a searchable text layer for English, Chinese, or Unicode-representable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

The formal contract is:

$$
\begin{aligned}
r_\alpha(\text{text prompt}) &\to x_{\text{prompt}}, \\
g_\beta(\text{uploaded page or glyph}) &\to x_{\text{prompt}}, \\
f_\theta(x_{\text{prompt}}, x_{\text{context}}) &\to y_{\text{page}}, \\
o_\psi(y_{\text{page}}) &\to \text{optional text layer} + \text{glyph-region metadata}.
\end{aligned}
$$

Here $r_\alpha$ is only a UI rasterizer, not the language model itself. The core model $f_\theta$ maps visual inputs to a rendered page image $y_{\text{page}}$. The optional reader $o_\psi$ may recover Unicode text where possible, but unencoded historical forms remain valid image regions.

A useful answer canvas should resemble a page excerpt from a book or corpus:

- a title region rendered as text image,

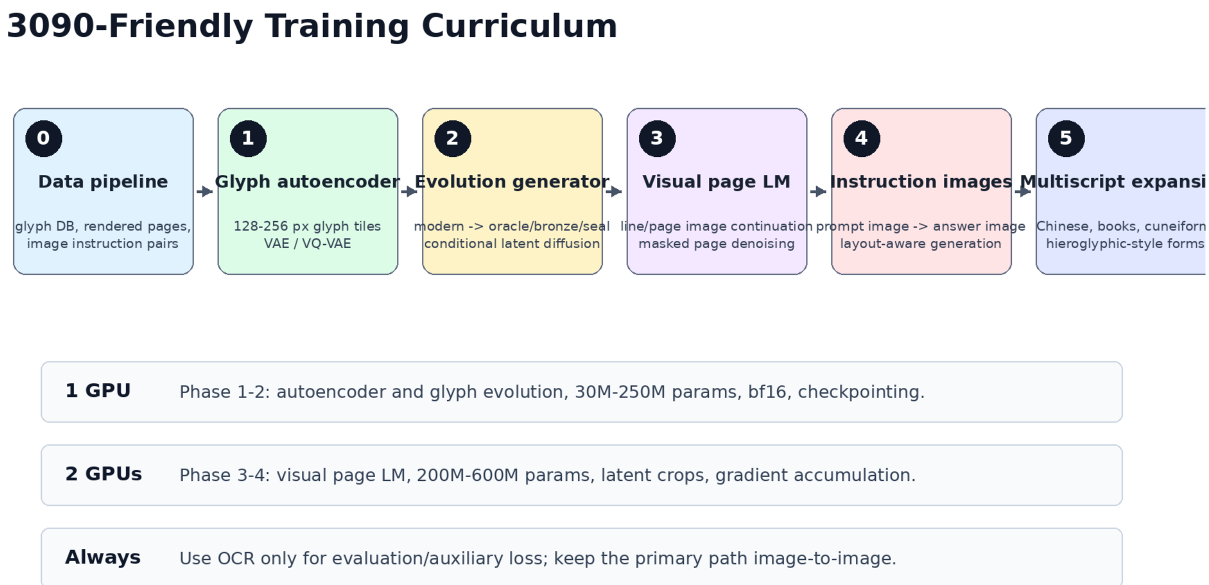


Figure 8: A staged curriculum that fits one or two RTX 3090 GPUs by starting with glyph tiles and latent generation before scaling to page-level image language modeling.

- one or more modern-language paragraphs, for example English and Chinese,
- a classical-style line or quotation when appropriate,
- glyph panels for forms that may not exist in computer fonts,
- a provenance strip with source IDs or nearest retrieved exemplars.

This schema is intentionally different from “text answer plus illustrations”. The generated page image is the primary output object, and the text layer is secondary.

# 9 3090-Friendly Curriculum

Figure 8 records the original page-model schedule. Measurements now support a different order:

1. Learn a cross-font foveal visual alphabet and audit missing-glyph coverage.
2. Train recurrent visual energy and conditional ink flow jointly on dense causal positions.
3. Reread generated endpoints during training and evaluate on a fixed external visual bank.
4. Train on autonomous model-induced visual trajectories; V6 completes this step and stabilizes ink but remains unreadable.
5. Train normalized context advantage on independent images and identity through differentiable sampled endpoints; V7 improves both signals but remains unreadable.

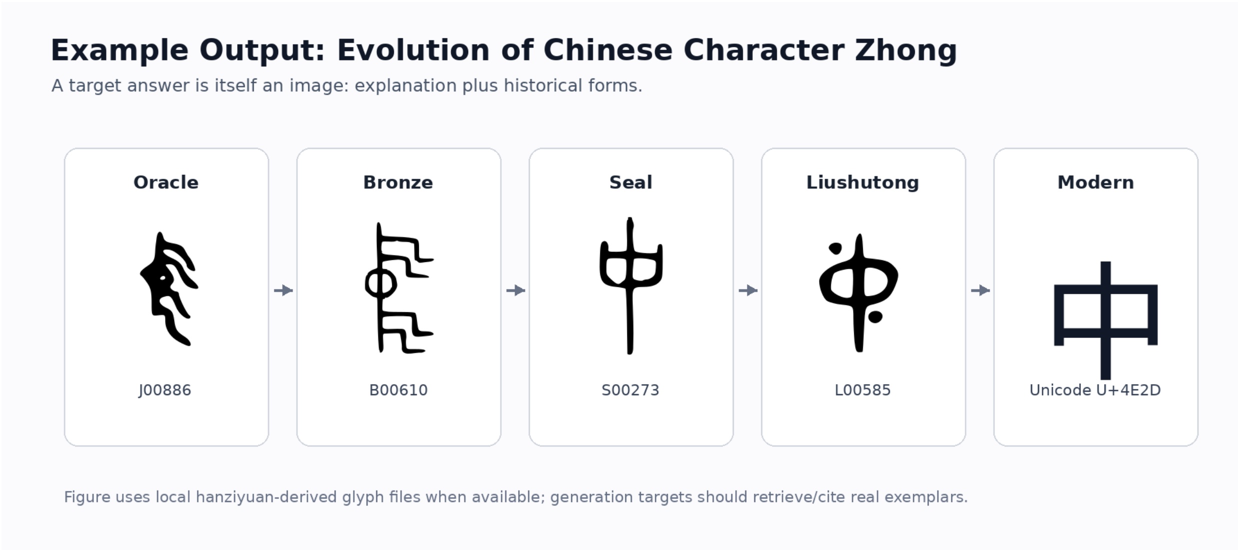


Figure 9: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

6. Train continuous proposal and next-retina flow alone; V15 passes random, last-only, unigram, context-use, and target-signal state gates.
7. Add residual multiscale causal visual memory; V16 replicates the gates with a seven-context proposal increase but does not beat the symbolic bigram.
8. Run a paired memory ablation on a newly preregistered bank before attributing the small V16 change to the memory module.
9. Condition a separate actuator on continuous visual plans; V17 passes isolated causal reread identity but fails topology and readability.
10. Decode a directly supervised spatial motor plan before stochastic refinement and require legible held-out forms.
11. Add provenance-gated historical panels and instruction images only after readable autonomous continuation.
12. Scale width and corpus size only after the fixed language and stability gates pass.

The 16.47M PVF V16 state model and 5.73M V17 actuator are deliberately small enough for one RTX 4090. Their measured peaks are 1.479 and 1.588 GiB, respectively, leaving substantial capacity headroom for controlled corrections. This is not an efficiency victory: quality remains below a symbolic bigram and pixel output is unreadable, so "more efficient than an LLM" remains an unsupported hypothesis.

## 10 Target Demonstration

The first demo should answer a rendered visual prompt such as: "Explain the evolution of the Chinese character yan." The target answer image should contain:

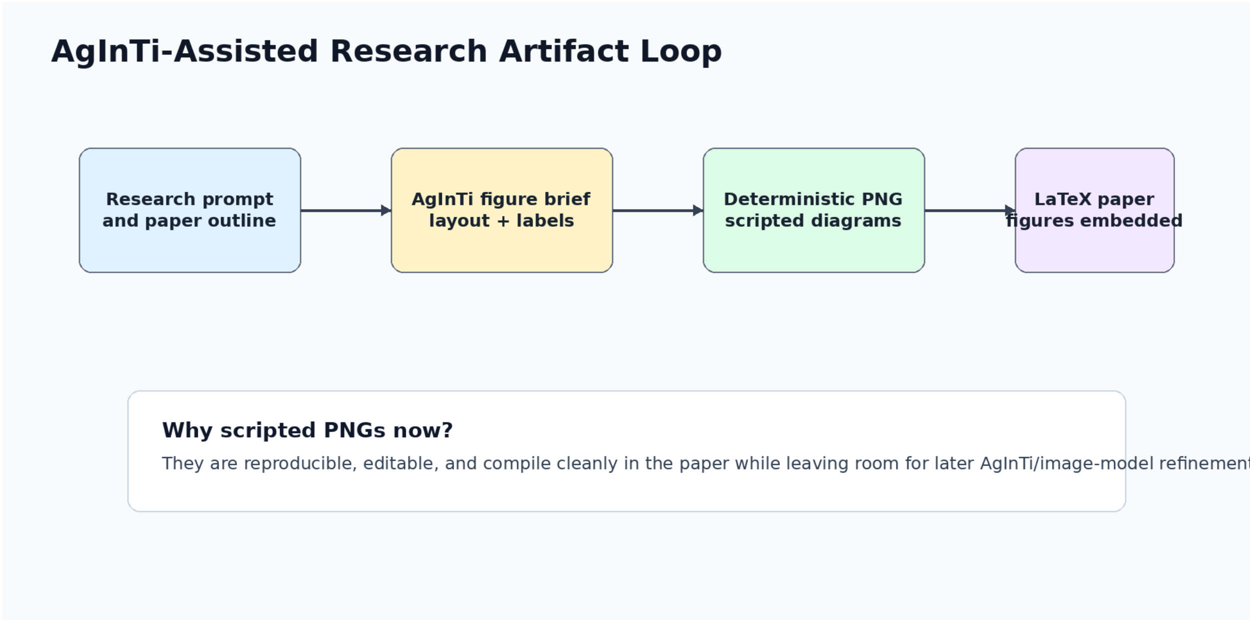


Figure 10: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

- a title and concise modern explanation,
- English and Chinese paragraphs rendered as page text,
- a classical-style line such as "yan is the visible trace of speech",
- a historical timeline with real oracle, bronze, seal, and modern glyph exemplars,
- stage labels and citations as rendered image text.

The existing Zhong figure in Figure 9 is a compact stage-timeline example; the Yan figures in Figures 1 and 7 show the fuller answer-page style. Together they test whether the model can combine language understanding, visual page composition, and historical script generation.

## 11 AgInTi Artifact Loop

The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

## 12 Evaluation

The model should be evaluated as both a language model and a visual generator:

- Fixed-bank top-1, top-5, and reciprocal rank for full context and a last-fixation ablation.
- Random-weight, chance, unigram, and symbolic bigram baselines on identical contexts.
- Cross-font retina-oracle retrieval, separated from recurrent language prediction.

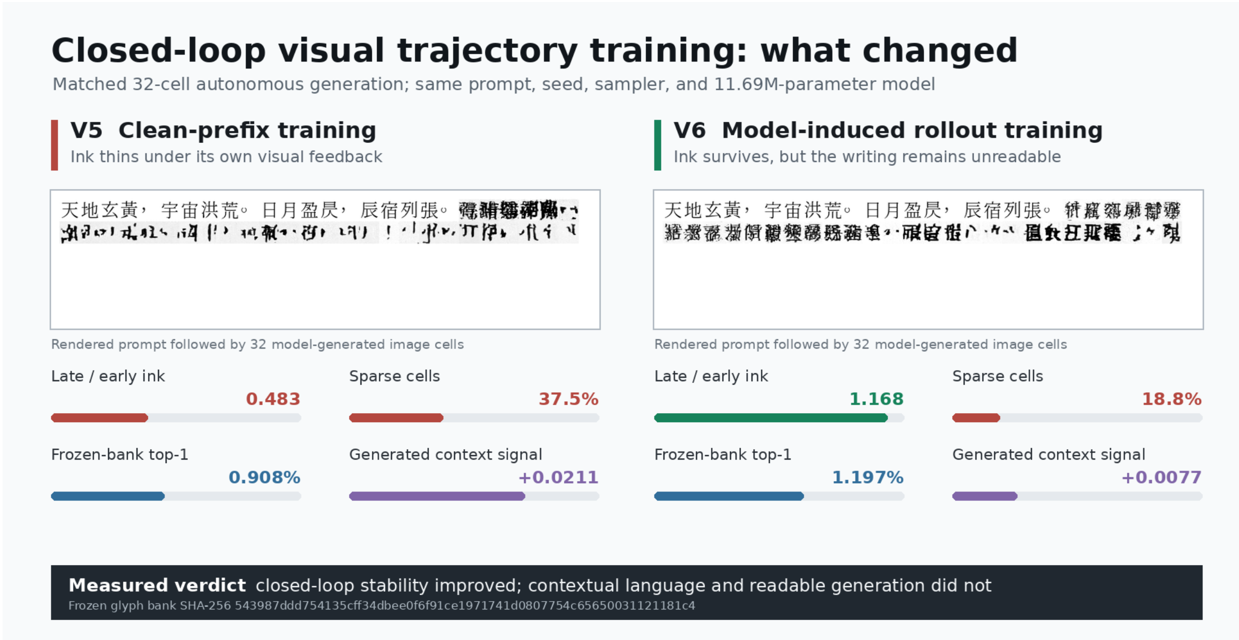


Figure 11: Matched autonomous evidence for V5 and V6. The prompt, seed, sampler, model size, and 32-cell horizon are fixed. Model-induced rollout training prevents the earlier loss of ink and halves the sparse-cell fraction, but neither continuation is readable and generated target signal weakens.

- Generated-image target similarity, pixel F1, ink occupancy, variance, and energy-reranked hit rate.
- Autonomous rollout length, throughput, VRAM, and blinded human readability.
- Student-boundary audit for token IDs, Unicode IDs, OCR, codebooks, labels, and external models.
- Later historical tasks: glyph-stage accuracy, attested-source provenance, and explicit synthetic-form labeling.


Figure 12: Matched autonomous evidence for V6 and selected V7 step 5,800. V7 improves calibrated retrieval and generated target signal while retaining stable ink, but visual inspection still finds dense unreadable pseudo-glyphs.

| System | Measured signal | Interpretation |
|---|---|---|
| Whole-page flow | velocity MSE 0.851 | Unreadable page texture; negative baseline |
| Global visual memory | top-1 2.23% over 2,016 | Above random, but 0/16 historical paraphrases |
| Causal InkStream | validation ink F1 0.639 | Autonomous repeated-grey collapse; negative baseline |
| RFLM V5 retina oracle | top-1 97.65% | Strong cross-font visual alphabet |
| RFLM V5 recurrent state | top-1 0.91% | Below last-only 1.36%, unigram 1.86%, and bigram 13.58% |
| RFLM V6 recurrent state | top-1 1.20% | Improved, but below last-only 1.69% and both language baselines |
| RFLM V6 pixel writer | context gain +0.0077 | Positive but weaker than V5's +0.0211 |
| RFLM V6 autonomous loop | ink ratio 1.168 | Stable character-scale marks remain unreadable |
| RFLM V7 recurrent state | top-1 2.31% | Above last-only 2.02% and unigram 1.86%, far below bigram 13.58% |
| RFLM V7 calibration | log-probability gain −0.2155 | Much better than V6's −0.9066, but full history still lowers mean target probability |
| RFLM V7 pixel writer | context cosine +0.0303 | Passes generated-signal gate; autonomous writing remains unreadable |

| PVF branch | Frozen signal | Interpretation |
|---|---|---|
| V14 proposal | top-1 2.52% | First proposal gate pass; above last-only 2.24% and unigram 1.73% |
| V14 state flow | top-1 2.35% | First stochastic state gate pass |
| V15 proposal | top-1 **5.87%** | Above last-only 4.42%, unigram 1.73%, and random 0.17%; context gain +0.0707 |
| V15 state flow | top-1 **3.41%** | Above last-only 2.68% and unigram; sampled context cosine gain +0.0805 |
| V16 proposal | top-1 **6.26%** | 112/1,788; above last-only 3.97% and V15 by seven contexts; context gain +0.0773 |
| V16 state flow | top-1 **3.69%** | Above last-only 3.24% and unigram; sampled context cosine gain +0.0795 |

Table 1: Frozen Predictive Visual Field results. V15 and V16 branches pass their visual-state gates but remain below the 13.14% symbolic bigram.

| V17 frozen actuator metric | Correct state | Shuffled state | Gate |
|---|---|---|---|
| Global identity top-1 | **58.594%** | 0.977% | pass |
| Target cosine | **0.71301** | 0.08612 | pass |
| Pixel F1 | **0.43849** | 0.28001 | fail (< 0.50) |
| Ink fraction | 0.15947 | 0.15967 | controlled |
| Human readability | rejected | rejected | fail |

Table 2: Untouched V17 evaluation with identical style and noise in each paired branch. Causal visual identity succeeds while readable topology fails.

The fixed evaluation contains 512 common Han candidates with four independent font views and 2,423 eligible held-out contexts. V6 and V7 use the identical bank SHA-256. Selected V7 raises full-context top-1 from 1.197% to 2.311% and top-5 from 3.219% to 5.613%, while the retina oracle remains near 98.27%. Full context now ranks above V7 last-only (2.022%) and unigram (1.857%) but remains far below symbolic bigram (13.578%). Its normalized target-log- probability gain is −0.2155, improved from a corrected V6 value of −0.9066 but still negative. Raw target-score gain stays positive in both models, showing that raw energy was a misleading metric. The formal acceptance result remains false, and retina-oracle accuracy cannot be cited as language accuracy.

PVF uses a separate frozen evaluation with the same 512 objects and four views for V14 through V16, bank SHA-256 `543987dd...1181c4`, and 1,788 eligible held-out contexts. V15 proposal top-1 is 5.872%, compared with 4.418% last-only, 1.734% unigram, 0.168% random dynamics, and 13.143% symbolic bigram. Proposal top-5 is 12.192%, and normalized context gain is +0.07069. The stochastic field obtains 3.412% top-1, 7.662% top-5, +0.03032 normalized context gain, and +0.08053 sampled context cosine gain. Random-dynamics sampled target cosine is 0.03180, versus 0.21576 for V15. The retina oracle is 98.546%.

V16 was selected at step 2,200 without reading that frozen bank and evaluated once. Its proposal obtains 6.264% top-1 (112/1,788), versus 3.971% last-only, 1.734% unigram, 0.224% random dynamics, 0.195% chance, and 13.143% symbolic bigram. Its proposal context gain is +0.07732. The stochastic field obtains 3.691% top-1 versus 3.244% last-only, normalized context gain +0.03579, sampled target cosine 0.19506, and sampled context cosine gain +0.07947. V16 adds seven correct proposal contexts over V15, while proposal top-5 and target cosine decline. This is directional evidence and not a statistically established memory advantage; paired predictions on a new frozen bank are required for attribution.

Thus the proposal and flow state gates are accepted, while the bigram language gate remains false. Separately, V17 accepts causal visual actuation and rejects readable topology. The combined result establishes that a small image-only student can learn causal language signal and that a small continuous state can control generated ink; it does not establish useful language. The next proof should isolate the multiscale memory contribution while a visual motor-plan decoder makes topology explicit. The final closed 32-cell loop must then remain readable. This order is also consistent with measured error accumulation in iterative diffusion chains [32].

# 13 Conclusion

The experiments show that merely replacing tokens with pixels is insufficient. A page flow can learn paper statistics, a causal writer can learn local ink, and a retina can learn character identity without acquiring language dynamics. RFLM contributes a strict test harness for ordered visual reading, direct ink flow, and autonomous rereading, but V6 and V7 reject exposure-bias correction and a monolithic language-plus-rendering writer as sufficient solutions. PVF then factors perception, causal language state, stochastic visual form, and actuation. V16's 6.26% proposal and 3.69% state-flow top-1, positive normalized history gains, and random/unigram/last-only controls replicate the accepted visual-state proof in this project. They show that causal language signal can be learned directly from writing images by a 16.47M model on one consumer GPU. V17 then reaches 58.59% frozen visual-identity retrieval versus 0.98% under shuffled intended states with only 5.73M trainable parameters and 1.588 GiB peak memory. This breaks the categorical claim that lightweight continuous visual intent cannot control generated writing. Its 0.438 pixel F1 and unreadable pseudo-characters reject the stronger claim. The seven-context V16 increase is too small to establish an architecture win, and the 13.14% bigram still rejects the causal core. The next attributable tests are paired multiscale-memory ablation and a spatial visual motor-plan actuator, followed only then by a closed write-reread loop. Historical etymology answers and broad instruction following remain downstream goals, not present capabilities, and no Qwen-8B parity or end-to-end efficiency claim is warranted.

# References


[1] Brown et al. Language Models are Few-Shot Learners. `https://arxiv.org/abs/2005.14165`

[2] OpenAI. GPT-4 Technical Report. `https://arxiv.org/abs/2303.08774`

[3] OpenAI. GPT-4o API model documentation. `https://developers.openai.com/api/docs/models/gpt-4o`

[4] OpenAI. Images and Vision API documentation. `https://platform.openai.com/docs/guides/images-vision`

[5] OpenAI. Image generation guide. `https://platform.openai.com/docs/guides/image-generation`

[6] Google. Gemini API image generation documentation. `https://ai.google.dev/gemini-api/docs/image-generation`

[7] Meta AI. The Llama 3 Herd of Models. `https://arxiv.org/abs/2407.21783`

[8] Kim et al. OCR-free Document Understanding Transformer. https://arxiv.org/abs/2111.15664

[9] Lee et al. Pix2Struct: Screenshot Parsing as Pretraining for Visual Language Understanding. https://arxiv.org/abs/2210.03347

[10] Rust et al. Language Modelling with Pixels. https://arxiv.org/abs/2207.06991

[11] Tai et al. PIXAR: Auto-Regressive Language Modeling in Pixel Space. https://arxiv.org/abs/2401.03321

[12] Chai et al. Autoregressive Pre-Training on Pixels and Texts. https://arxiv.org/abs/2404.10710

[13] Clark et al. CANINE: Pre-training an Efficient Tokenization-Free Encoder. https://arxiv.org/abs/2103.06874

[14] Xue et al. ByT5: Towards a Token-Free Future. https://arxiv.org/abs/2105.13626

[15] Yu et al. MEGABYTE: Predicting Million-byte Sequences with Multiscale Transformers. https://arxiv.org/abs/2305.07185

[16] Austin et al. Structured Denoising Diffusion Models in Discrete State-Spaces. https://arxiv.org/abs/2107.03006

[17] Li et al. Diffusion-LM Improves Controllable Text Generation. https://arxiv.org/abs/2205.14217

[18] Lou et al. Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution. https://arxiv.org/abs/2310.16834

[19] Nie et al. Large Language Diffusion Models. https://arxiv.org/abs/2502.09992

[20] Chang et al. MaskGIT: Masked Generative Image Transformer. https://arxiv.org/abs/2202.04200

[21] Rombach et al. High-Resolution Image Synthesis with Latent Diffusion Models. https://arxiv.org/abs/2112.10752

[22] Peebles and Xie. Scalable Diffusion Models with Transformers. https://arxiv.org/abs/2212.09748

[23] Lipman et al. Flow Matching for Generative Modeling. https://arxiv.org/abs/2210.02747

[24] Chen and Lipman. Riemannian Flow Matching on General Geometries. https://arxiv.org/abs/2302.03660

[25] Mathieu and Nickel. Riemannian Continuous Normalizing Flows. https://arxiv.org/abs/2006.10605

[26] Assran et al. Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. https://arxiv.org/abs/2301.08243

[27] Velarde and Parra. Recurrent Joint Embedding Predictive Architecture with Recurrent Forward Propagation Learning. https://arxiv.org/abs/2411.16695

[28] Chen et al. Denoising with a Joint-Embedding Predictive Architecture. https://arxiv.org/abs/2410.03755

[29] Hu et al. ELF: Embedded Language Flows. https://arxiv.org/abs/2605.10938

[30] Ross, Gordon, and Bagnell. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning. https://proceedings.mlr.press/v15/ross11a.html

[31] Bengio et al. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. https://papers.nips.cc/paper_files/paper/2015/hash/e995f98d56967d946471af29d7bf99f1-Abstract.html

[32] Ning et al. Input Perturbation Reduces Exposure Bias in Diffusion Models. https://proceedings.mlr.press/v202/ning23a.html

[33] Yan et al. Eye movement guidance in Chinese reading: Is there a preferred viewing location? *Vision Research*, 51(10), 2011. https://doi.org/10.1016/j.visres.2011.03.004